

Online Appendix

From Cradle to Consequence: The Dynamic Impact of Parental Behaviors on Teenage Outcomes Among Fragile Families

Danny Turkson

University of North Carolina, Chapel Hill

This online appendix reports some descriptives and the full set of estimation tables underlying the results discussed in the main paper, including the endogenous parental behavior equations, the child cognitive achievement equation, and the teenage outcome equations, each estimated with and without unobserved heterogeneity (No UH vs. With UH). Tables are numbered independently from the main paper (OA.1, OA.2, ...) and are cross-referenced from the main text where the corresponding results are discussed. The results for the Alternative Models are also presented in this online appendix.

1 Descriptives

Table OA.1: **Distribution of endogenous variables across wave**

Variable	Pattern	%	Variable	Pattern	%
Mom Employed	111111	44.21	Mom Welfare Receipt	000000	47.00
	111110	7.93		100000	10.65
	101111	3.84		110000	3.84
	111101	3.47		000010	2.91
	011111	2.91		111000	2.91
	000000	2.79		010000	2.35
	Others	34.86		Others	30.34
Mom Alcohol Use	011111	16.59	Mom Smokes	000000	61.24
	000000	14.86		111111	8.11
	001111	10.84		011111	3.22
	000001	5.14		110111	2.72
	000011	5.02		000100	1.80
	000111	4.71		000001	1.30
	111111	4.58		Others	21.61
Mom Marijuana Use	Others	38.27	Mom Ever Convicted	000000	88.79
	000000	90.90		001111	5.08
	000001	2.54		000001	2.48
	000011	1.05		000111	1.80
	000010	0.99		000011	1.24
	Others	4.52		111111	0.43
				011111	0.19
Biological Dad Present	000000	24.83	Another Dad Present	000000	70.53
	111111	23.72		000001	19.38
	100000	7.99		000011	3.96
	110000	6.25		000010	1.61
	111110	5.57		000111	1.24
	111000	5.02		000101	0.68
	111100	4.89		Others	2.60
Any Dad Figure Present	Others	21.73	Dad Ever in Jail	000000	48.92
	111111	25.39		111111	26.07
	000000	13.87		001111	10.77
	000001	6.87		011111	5.02
	111110	4.52		000011	4.77
	100000	4.15		000111	2.72
	111101	3.59		000001	1.73
	110000	2.97			
	Others	38.64			

Note: Patterns presented in order of prevalence

Table OA.2: Summary statistics: Exogenous variables (N = 1615)

Wave	1	2	3	4	5	6
Child's age	Birth	Year 1	Year 3	Year 5	Year 9	Year15
Family characteristics						
Mom age (average)	25.3	26.5	28.2	30.3	34.4	40.4
Mom Education						
Less High Sch	0.28	0.24	0.22	0.20	0.17	0.15
High Sch	0.31	0.30	0.28	0.25	0.20	0.18
Some College	0.28	0.32	0.34	0.39	0.44	0.44
College or Grad	0.13	0.14	0.15	0.16	0.19	0.23
Lives in Public Housing	0.11	0.13	0.13	0.05	0.05	0.08
Grandparent lives in HH	0.27	0.20	0.16	0.12	0.11	0.72
Mom average non-wage income (\$)		2,094	1,771	1,477	3,406	3,894
Number of siblings (average)		0.11	0.42	0.54	0.54	0.73
Child characteristics						
WJ English score					25.91	
WJ Math score					32.40	
Time spent on school activity						
Never						0.64
Once per month						0.15
Once per week						0.12
More than once per week						0.09
Time spent on religious activity						
Never						0.59
Once per month						0.15
Once per week						0.18
More than once per week						0.08
Disobedient at school						0.24
Cheats						0.28
Bullies' peers						0.17
Attend public school					0.63	
Diagnosed ADHD					0.11	
BMI (average)						23.8
Peer characteristics						
Peers likely to graduate from college						0.48
Peers study						
None of time						0.34
Most times						0.66
Peers smoke						0.17
Peers drink						0.26
Peers use marijuana						0.32

Note: Non-wage income is inflation-adjusted and reported in year 2000 dollars values.

2 Full Estimation Tables

Table OA.3: Estimation of Mother's Marital Status: No UH vs. With UH

Variable	Married/Cohabiting with Biological Dad				Married/Cohabiting with Another Man			
	No UH		UH		No UH		UH	
	Coefficient	Std. Dev.	Coefficient	Std. Dev.	Coefficient	Std. Dev.	Coefficient	Std. Dev.
with biodad _{t-1}	3.13081***	0.0814	51.56484***	0.3767	-1.40387***	0.1956	-0.72143***	0.2402
with anotherman _{t-1}	-2.15552**	1.0111	-21.62795***	1.3265	1.95446***	0.2150	1.94372***	0.2425
employ _{t-1}	-0.36929***	0.0884	-0.58165*	0.3050	0.01306	0.1364	0.00548	0.1496
welfare _{t-1}	-0.30174***	0.0986	-0.98966***	0.3398	-0.02223	0.1455	-0.02681	0.1565
drink _{t-1}	-0.05042	0.0752	0.23488	0.3664	0.06117	0.1134	0.01717	0.1518
smoke _{t-1}	-0.27307***	0.0858	-0.81987*	0.4701	-0.11008	0.1243	-0.28751	0.2370
marijuana _{t-1}	-0.20249	0.2815	-0.09505	0.8394	0.66221**	0.2578	0.59244**	0.2781
convicted _{t-1}	-0.93703***	0.3579	-1.76087	1.4014	-0.35431	0.3701	-0.49305	0.3756
cognitive score _{t-1}	0.06497**	0.0315	0.28291**	0.1207	-0.06415	0.0591	-0.07501	0.0632
dad in jail _{t-1}	-0.25102	0.1889	-0.54503	0.7280	0.02055	0.1677	0.03948	0.1793
age	0.07369**	0.0336	0.34794**	0.1492	0.11256	0.0710	0.13700*	0.0744
age ²	-0.00066	0.0005	-0.00325*	0.0018	-0.00182*	0.0009	-0.00200**	0.0010
white	0.67449***	0.1058	1.95088***	0.4992	0.90298***	0.1618	1.05505***	0.1837
hispanic	0.50011***	0.0915	1.07044***	0.3487	0.38049***	0.1463	0.36875**	0.1587
other race	0.58581***	0.1917	2.10452**	0.8828	-0.25367	0.3133	-0.20471	0.3558
less than high sch	-0.02980	0.1076	-0.27796	0.3581	0.01552	0.1738	0.09877	0.1845
some college	-0.09701	0.0882	-0.19992	0.3165	0.18014	0.1388	0.21163	0.1454
graduate	0.37275***	0.1255	19.80943***	0.8505	0.55396***	0.1785	0.64392***	0.2105
very good health	0.08308	0.0882	0.29463	0.3175	0.10078	0.1442	0.05980	0.1529
good health	-0.04411	0.0935	0.11410	0.3276	0.01863	0.1456	-0.06191	0.1543
fair/poor health	-0.22440*	0.1196	-0.37090	0.3718	-0.03454	0.1684	-0.11353	0.1786
no. of kids	0.15212*	0.0876	0.42311*	0.2498	0.42465***	0.1159	0.50884***	0.1250
no. of kids ²	-0.02047	0.0132	-0.06807**	0.0303	-0.02677*	0.0143	-0.03822**	0.0155
pregnant/new birth	-0.04094	0.0902	-0.26923	0.3063	0.26876*	0.1382	0.24410	0.1504
grandparent present	0.26368***	0.0950	4.96978***	0.6676	1.83026***	0.1132	2.19715***	0.1530
public housing	-0.38740***	0.1342	-1.31720***	0.3595	-0.12375	0.1905	-0.17430	0.2020
non-wage income	0.00520	0.0059	0.03909**	0.0154	-0.01792	0.0122	-0.01836	0.0131
county fraction married	0.59941	0.5557	0.02841	1.6753	0.27695	0.7162	0.14652	0.7476
city unemployment rate	0.01061	0.0224	0.18594**	0.0799	-0.05856*	0.0343	-0.05429	0.0357
county % on public assistance	0.44467	0.6975	3.93350	2.3171	-3.13693**	1.2283	-3.15987**	1.3498
state alcohol beer tax	0.19547	0.2594	-0.98879	0.8381	0.49387*	0.2813	0.48897	0.3042
state cigarettes tax	0.10544*	0.0551	0.06104	0.1722	-0.17233**	0.0695	-0.17859**	0.0743
state marijuana law	0.19376*	0.0987	0.16878	0.3103	0.00828	0.1268	0.01638	0.1298
county violent crime rate	-0.02365	0.1145	-0.12483	0.3624	0.31192*	0.1691	0.25848	0.1807
time trend	0.00710	0.1311	-0.74966	0.5734	1.08001***	0.3348	1.08265***	0.3488
time trend ²	-0.02554*	0.0153	0.00877	0.0639	-0.07328**	0.0339	-0.07225**	0.0352
time trend ³	0.00118**	0.0005	0.00005	0.0022	0.00184	0.0011	0.00180	0.0011
constant	-3.84746***	0.7648	-3.95428	2.8591	-9.66771***	1.4982	-9.54927***	1.8594
Permanent UH								
type 2			-0.89989	0.5442			-0.03010	0.2217
type 3			0.30977	0.5593			0.01695	0.2558
type 4			-1.03506	0.9234			0.28722	0.3896
Time-varying UH								
type 2			-50.65416***	0.3954			-0.85512	1.1932
type 3			-51.93086***	0.5125			-1.03376	1.2892
type 4			-75.97527***	0.6561			-1.22211	1.2273

Note: Permanent and Time-varying UH are relative to type 1, which are normalized to 0. Coefficients reported are log odds. The reference category is Single (No Dad Present). *** p<0.01, ** p<0.05, * p<0.1.

Table OA.4: Estimation of Mother's Employment: No UH vs. With UH

Variable	No UH		UH	
	Coefficient	Std. Dev.	Coefficient	Std. Dev.
with biodad _{t-1}	0.00093	0.0670	0.04772	0.0794
with anotherman _{t-1}	0.05136	0.1895	0.01360	0.2063
employ _{t-1}	1.82740***	0.0782	1.76903***	0.1140
welfare _{t-1}	-0.17779**	0.0809	-0.11387	0.0960
drink _{t-1}	0.20126***	0.0696	-0.14767	0.1065
smoke _{t-1}	-0.14773*	0.0773	0.17858	0.1534
marijuana _{t-1}	-0.35811*	0.2071	-0.70073***	0.2344
convicted _{t-1}	-0.01089	0.2763	-0.06092	0.3286
dad in jail _{t-1}	0.09744	0.1121	0.02092	0.1498
cognitive score _{t-1}	0.15352***	0.0298	0.12577***	0.0367
age	-0.02903***	0.0058	-0.02975***	0.0070
white	-0.44052***	0.0970	-0.43797***	0.1150
hispanic	-0.40643***	0.0868	-0.49220***	0.1069
other race	-0.36883*	0.1953	-0.45387*	0.2384
less than high sch	-0.26556***	0.0906	-0.34273***	0.1028
some college	0.26058***	0.0808	0.21822**	0.0932
graduate	0.34922***	0.1172	0.38480***	0.1359
very good health	-0.06201	0.0799	-0.07021	0.0859
good health	-0.11291	0.0839	-0.13380	0.0898
fair/poor health	-0.42609***	0.0992	-0.46371***	0.1079
no. of kids	-0.06131**	0.0233	-0.07197***	0.0259
pregnant/new birth	-0.36115***	0.0788	-0.37083***	0.0841
grandparent present	0.17526**	0.0835	0.19086**	0.0924
public housing	-0.14612	0.1091	-0.14364	0.1153
non-wage income	0.00458	0.0032	0.00480	0.0035
county fraction married	-1.45999***	0.5101	-1.38532**	0.5516
city unemployment rate	0.06019**	0.0220	0.05814**	0.0232
county % on public assistance	-0.54069	0.5363	-0.22374	0.5816
state alcohol beer tax	0.05818	0.2351	0.11457	0.2539
state cigarettes tax	-0.04915	0.0529	-0.04206	0.0569
state marijuana law	-0.16861*	0.0980	-0.16058	0.1084
county violent crime rate	-0.05918	0.1013	-0.04427	0.1137
time trend	0.29154***	0.1039	0.44619***	0.1165
time trend ²	-0.03185**	0.0126	-0.04638***	0.0138
time trend ³	0.00100**	0.0004	0.00143***	0.0005
constant	-0.05679	0.5057	-0.11546	0.6017
Permanent UH				
type 2			0.76970***	0.2601
type 3			-0.48820**	0.1951
type 4			0.69357***	0.2663
Time-varying UH				
type 2			-0.59945***	0.1957
type 3			1.05269	0.7003
type 4			-0.02065	0.2229

Note: Permanent and Time-varying UH are relative to type 1, which are normalized to 0.

Coefficients reported are log odds. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table OA.5: Estimation of Mother's Welfare Receipt: No UH vs. With UH

Variable	No UH		UH	
	Coefficient	Std. Dev.	Coefficient	Std. Dev.
with biodad _{t-1}	-0.53837***	0.0772	-0.70897***	0.0936
with anotherman _{t-1}	-0.28392	0.2395	-0.33784	0.2641
employ _{t-1}	-0.18089**	0.0878	-0.16845	0.1147
welfare _{t-1}	1.47087***	0.0781	1.61046***	0.0931
drink _{t-1}	0.01659	0.0805	0.14247	0.1357
smoke _{t-1}	0.40173***	0.0814	0.16657	0.1544
marijuana _{t-1}	0.31957	0.2126	0.50994**	0.2581
convicted _{t-1}	0.35807	0.2765	0.42761	0.3257
cognitive score _{t-1}	-0.13074***	0.0324	-0.13260***	0.0404
dad in jail _{t-1}	0.35730***	0.1140	0.00895	0.1729
age	-0.11075***	0.0381	-0.14720***	0.0457
age ²	0.00116**	0.0006	0.00164**	0.0007
white	-0.37998***	0.1232	-0.46074***	0.1421
hispanic	-0.41016***	0.1042	-0.42572***	0.1185
other race	-0.26317	0.2476	-0.26397	0.2878
less than high sch	0.05969	0.0940	0.12407	0.1112
some college	-0.23800***	0.0883	-0.22003**	0.1009
graduate	-1.13715***	0.2087	-1.18441***	0.2195
very good health	-0.00745	0.0946	-0.03400	0.1074
good health	0.01847	0.0953	0.01043	0.1098
fair/poor health	0.50358***	0.1069	0.57370***	0.1250
no. of kids	0.04033	0.0265	0.05666*	0.0305
pregnant/new birth	0.35847***	0.0856	0.40500***	0.0990
grandparent present	0.00792	0.0933	-0.09573	0.1072
public housing	0.50733***	0.1025	0.59580***	0.1275
non-wage income	-0.00202	0.0024	-0.00348	0.0032
county fraction married	-1.99318***	0.6569	-2.28991***	0.7437
city unemployment rate	0.03687	0.0264	0.04366	0.0304
county % on public assistance	2.39535***	0.6010	2.42689***	0.7331
state alcohol beer tax	-1.25472***	0.3772	-1.47023***	0.4180
state cigarettes tax	-0.05430	0.0665	-0.06980	0.0763
state marijuana law	0.53467***	0.1180	0.57896***	0.1329
county violent crime rate	-0.02232	0.1114	-0.03350	0.1311
time trend	-0.22990**	0.1168	-0.27875**	0.1390
time trend ²	0.02845**	0.0142	0.03226*	0.0165
time trend ³	-0.00111**	0.0005	-0.00119**	0.0006
constant	2.80217***	0.7579	3.23209***	0.9142
Permanent UH				
type 2			-0.16874	0.2997
type 3			0.53918***	0.1950
type 4			-0.02914	0.2935
Time-varying UH				
type 2			0.62371***	0.1923
type 3			-15.49415***	0.0000
type 4			1.37479***	0.2128

Note: Permanent and Time-varying UH are relative to type 1, which are normalized to 0.
Coefficients reported are log odds. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table OA.6: Estimation of Mother's Drinking Behavior: No UH vs. With UH

Variable	No UH		UH	
	Coefficient	Std. Dev.	Coefficient	Std. Dev.
with biodad _{t-1}	0.05379	0.0595	0.22459**	0.1008
with anotherman _{t-1}	-0.05531	0.1713	-0.14840	0.2228
employ _{t-1}	0.19183***	0.0700	-0.17416	0.1081
welfare _{t-1}	-0.08938	0.0676	-0.02658	0.1126
drink _{t-1}	1.79159***	0.0698	1.08761***	0.1519
smoke _{t-1}	0.33554***	0.0684	-0.32061**	0.1470
marijuana _{t-1}	0.85409***	0.2488	0.39599	0.3069
convicted _{t-1}	-0.04771	0.2345	-0.18873	0.4337
cognitive score _{t-1}	0.08357***	0.0257	-0.00823	0.0589
age	0.03124	0.0248	0.11659***	0.0321
age ²	-0.00082**	0.0004	-0.00197***	0.0005
white	0.22978***	0.0891	0.21568	0.1611
hispanic	-0.04012	0.0772	-0.34658**	0.1439
other race	-0.39723**	0.1596	-0.80751***	0.2975
less than high sch	-0.19172**	0.0825	-0.28027**	0.1215
some college	0.10664	0.0707	0.11301	0.1201
graduate	0.43404***	0.1077	0.57379***	0.1663
very good health	0.10727	0.0657	0.10498	0.0853
good health	0.16987**	0.0721	0.16685*	0.0925
fair/poor health	0.20166**	0.0924	0.24215**	0.1208
no. of kids	-0.01002	0.0216	-0.02891	0.0295
pregnant/new birth	-0.42833***	0.0707	-0.49442***	0.0890
grandparent present	-0.06510	0.0693	-0.09534	0.0857
public housing	0.04752	0.0976	0.10541	0.1118
non-wage income	0.00155	0.0022	-0.00024	0.0022
county fraction married	0.11555	0.4487	0.11024	0.5891
city unemployment rate	0.05203***	0.0193	0.06431**	0.0261
county % on public assistance	-1.58497***	0.4858	-1.35581**	0.6449
state alcohol beer tax	-0.18944	0.2097	-0.08709	0.3058
state cigarettes tax	0.01160	0.0458	0.04279	0.0594
state marijuana law	0.14988*	0.0847	0.29093**	0.1284
county violent crime rate	-0.04189	0.0863	-0.07885	0.1200
time trend	0.20974**	0.0943	0.59416***	0.1185
time trend ²	-0.02140*	0.0114	-0.05632***	0.0139
time trend ³	0.00071*	0.0004	0.00173***	0.0005
constant	-2.48953***	0.5577	-4.91037***	0.9020
Permanent UH				
type 2			2.25064***	0.1890
type 3			1.20664***	0.3162
type 4			3.16485***	0.3976
Time-varying UH				
type 2			-0.16723	0.1796
type 3			-0.84571**	0.3298
type 4			-0.16395	0.1828

Note: Permanent and Time-varying UH are relative to type 1, which are normalized to 0.

Coefficients reported are log odds. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table OA.7: Estimation of Mother's Smoking Equation: No UH vs With UH

Variable	No UH		UH	
	Coefficient	Std. Dev.	Coefficient	Std. Dev.
with biodad _{t-1}	-0.20705***	0.0772	-0.15510	0.1532
with anotherman _{t-1}	0.03808	0.2256	0.12727	0.3715
employ _{t-1}	-0.14768*	0.0879	-0.12373	0.1686
welfare _{t-1}	0.25524***	0.0886	0.19079	0.1354
drink _{t-1}	0.17129**	0.0773	-0.20477	0.1535
smoke _{t-1}	3.42961***	0.0979	1.33221***	0.2208
marijuana _{t-1}	0.64847***	0.2116	0.29584	0.3221
convicted _{t-1}	0.50552*	0.2609	0.64433	0.4108
cognitive score _{t-1}	-0.01744	0.0321	-0.04469	0.0755
age	-0.03140***	0.0067	-0.02051	0.0192
white	0.49344***	0.1031	0.42251	0.2883
hispanic	-0.21004**	0.1039	-0.60194**	0.2656
other race	-0.09251	0.1963	-0.55148	0.5337
less than high sch	0.10147	0.1042	0.24271	0.2148
some college	-0.05497	0.0890	0.18679	0.2187
graduate	-0.66315***	0.1395	-1.02406***	0.3829
very good health	0.32021***	0.0966	0.38657***	0.1480
good health	0.32930***	0.0972	0.45654***	0.1640
fair/poor health	0.45859***	0.1150	0.77312***	0.1889
no. of kids	0.03042	0.0256	0.09266*	0.0516
pregnant/new birth	-0.48158***	0.1018	-0.71323***	0.1310
grandparent present	0.09871	0.0908	-0.07748	0.1368
public housing	0.04622	0.1376	0.00205	0.1833
non-wage income	0.00211	0.0019	0.00210	0.0028
county fraction married	0.36286	0.5901	-0.17397	0.9329
city unemployment rate	-0.08576***	0.0244	-0.11280***	0.0382
county % on public assistance	1.23192**	0.5729	0.73084	0.9609
state alcohol beer tax	-0.14308	0.2783	-0.25335	0.4612
state cigarettes tax	-0.09430	0.0631	-0.10303	0.1052
state marijuana law	-0.03213	0.1073	0.00153	0.2314
county violent crime rate	0.11655	0.1102	0.05419	0.2134
time trend	-0.01329	0.1170	0.11267	0.1591
time trend ²	0.01350	0.0141	0.01414	0.0188
time trend ³	-0.00074	0.0005	-0.00106	0.0007
constant	-1.30343**	0.5723	-4.68432***	1.4705
Permanent UH				
type 2			2.67761***	0.7379
type 3			5.70484***	0.6461
type 4			6.24595***	0.7230
Time-varying UH				
type 2			-0.41674	0.2706
type 3			-1.35429***	0.4817
type 4			0.40618	0.2853

Note: Permanent and Time-varying UH are relative to type 1, which are normalized to 0.

Coefficients reported are log odds. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table OA.8: Estimation of Mother's Marijuana Use: No UH vs With UH

Variable	No UH		UH	
	Coefficient	Std. Dev.	Coefficient	Std. Dev.
with biodad _{t-1}	-0.20511	0.1723	-0.14869	0.1893
with anotherman _{t-1}	0.63201**	0.3142	0.54090*	0.3231
employ _{t-1}	-0.31410	0.1991	-0.69964***	0.2352
welfare _{t-1}	0.08322	0.1605	0.13366	0.1813
drink _{t-1}	0.61055***	0.1763	-0.10696	0.2312
smoke _{t-1}	1.10912***	0.1757	0.33299	0.2682
marijuana _{t-1}	3.03465***	0.2875	2.88030***	0.3149
convicted _{t-1}	0.68135*	0.3557	0.56536	0.4305
cognitive score _{t-1}	0.05815	0.0648	-0.03460	0.0771
age	-0.02019	0.0165	-0.01566	0.0184
white	-0.30499	0.2361	-0.30852	0.2690
hispanic	-0.69129**	0.2926	-0.89694***	0.3006
other race	-0.99047	0.9185	-1.15057	0.9303
less than high sch	-0.21674	0.2275	-0.34536	0.2484
some college	-0.01292	0.1977	-0.06498	0.2235
graduate	0.37192	0.2935	0.25092	0.3215
very good health	0.28946	0.2112	0.28648	0.2182
good health	0.26800	0.2226	0.22756	0.2294
fair/poor health	0.73536***	0.2388	0.80082***	0.2422
no. of kids	-0.39543**	0.1724	-0.44507**	0.1850
no. of kids ²	0.03903*	0.0231	0.04690*	0.0242
pregnant/new birth	0.22172	0.2064	0.21007	0.2114
grandparent present	-0.26362	0.1935	-0.33962*	0.2001
public housing	0.02164	0.2602	0.12484	0.2785
non-wage income	0.00347*	0.0018	0.00418	0.0027
county fraction married	-1.63337	1.0951	-1.71174	1.1481
city unemployment rate	0.01342	0.0522	0.01926	0.0551
county % on public assistance	0.91282	1.2910	0.98291	1.4532
state alcohol beer tax	-1.01947	0.6528	-1.06387	0.7555
state cigarettes tax	-0.20508	0.1336	-0.20491	0.1510
state marijuana law	0.54441**	0.2359	0.60338**	0.2621
county violent crime rate	0.25492	0.2302	0.24290	0.2450
time trend	-0.19718	0.2500	0.07242	0.2684
time trend ²	0.02731	0.0303	0.00386	0.0322
time trend ³	-0.00082	0.0011	-0.00015	0.0011
constant	-2.64340**	1.2464	-4.38672***	1.5147
Permanent UH				
type 2			2.74536***	0.6963
type 3			2.23788***	0.6424
type 4			3.87321***	0.6381
Time-varying UH				
type 2			-0.19903	0.4462
type 3			-0.91870	0.6651
type 4			0.64661	0.4055

Note: Permanent and Time-varying UH are relative to type 1, which are normalized to 0.

Coefficients reported are log odds. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table OA.9: Estimation of Mother's Conviction: No UH vs With UH

Variable	No UH		UH	
	Coefficient	Std. Dev.	Coefficient	Std. Dev.
with biodad _{t-1}	-0.39631*	0.2119	-0.45988**	0.2146
with anotherman _{t-1}	-0.91555	0.5635	-1.09612*	0.5987
employ _{t-1}	-0.01172	0.2166	-0.14783	0.2508
welfare _{t-1}	0.45984**	0.2062	0.39686*	0.2239
drink _{t-1}	0.35181*	0.2052	0.34784	0.2764
smoke _{t-1}	0.78468***	0.1872	0.33504	0.3123
marijuana _{t-1}	0.38547	0.3647	0.36237	0.4569
convicted _{t-1}	2.59098***	0.3323	2.84570***	0.4088
cognitive score _{t-1}	-0.05688	0.0773	-0.06209	0.0843
age	0.31684**	0.1318	0.31485**	0.1354
age ²	-0.00495**	0.0019	-0.00491**	0.0020
white	0.12805	0.2240	0.09939	0.2443
hispanic	-0.17260	0.2691	-0.23958	0.2801
other race	0.01230	0.4296	-0.02077	0.4898
less than high sch	-0.17878	0.2629	-0.19964	0.2787
some college	-0.01238	0.2147	-0.02852	0.2260
graduate	-0.79002*	0.4532	-0.89194*	0.4573
very good health	0.05503	0.2614	0.00648	0.2767
good health	0.13161	0.2606	0.07855	0.2760
fair/poor health	0.69426**	0.2786	0.70439**	0.2942
no. of kids	-0.13682*	0.0722	-0.12773*	0.0732
pregnant/new birth	0.42865**	0.2168	0.38331*	0.2255
grandparent present	-0.06043	0.2495	-0.13273	0.2511
public housing	0.10470	0.3123	0.18716	0.3366
non-wage income	0.00428**	0.0020	0.00409	0.0025
county fraction married	0.58550	1.4456	0.61995	1.5165
city unemployment rate	-0.04097	0.0609	-0.03627	0.0620
county % on public assistance	0.42821	1.5947	0.23800	1.7509
state alcohol beer tax	0.16641	0.5024	0.06278	0.5416
state cigarettes tax	-0.22935*	0.1358	-0.27066*	0.1534
state marijuana law	-0.08226	0.2816	-0.00873	0.2930
county violent crime rate	-0.52468*	0.2942	-0.54218*	0.3065
time trend	0.63991	0.4950	0.67922	0.5061
time trend ²	-0.05968	0.0533	-0.06223	0.0547
time trend ³	0.00190	0.0017	0.00200	0.0018
constant	-10.41185***	2.7265	-10.34567***	2.8582
Permanent UH				
type 2			-0.24524	0.5421
type 3			0.29720	0.4594
type 4			0.79973	0.5198
Time-varying UH				
type 2			-0.38707	0.7542
type 3			-56.05953***	0.0000
type 4			1.16219**	0.5300

Note: Permanent and Time-varying UH are relative to type 1, which are normalized to 0.

Coefficients reported are log odds. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table OA.10: Estimation of Dad in Jail: No UH vs With UH

Variable	No UH		UH	
	Coefficient	Std. Dev.	Coefficient	Std. Dev.
in-jail _{t-1}	2.48022***	0.1412	2.90853***	0.1884
with bio-mom _{t-1}	-0.72601***	0.1252	-0.87040***	0.1342
employ _{t-1}	-0.35174***	0.1161	-0.24696*	0.1275
welfare _{t-1}	0.27419	0.3466	0.46227	0.3941
drink _{t-1}	-0.25307**	0.1148	-0.37387***	0.1295
smoke _{t-1}	0.49824***	0.1072	0.53134***	0.1267
cognitive score _{t-1}	-0.08445*	0.0446	-0.13791**	0.0552
white	-0.40756**	0.1716	-0.54061***	0.1828
hispanic	-0.22650*	0.1229	-0.23949*	0.1401
other race	-0.35022	0.4456	-0.43398	0.4810
less than high sch	0.28200**	0.1098	0.35364***	0.1249
some college	-0.24848*	0.1338	-0.25616*	0.1505
graduate	-1.48819***	0.4915	-1.47951***	0.4904
very good health	-0.28908**	0.1128	-0.34588***	0.1254
good health	-0.32022**	0.1335	-0.36840**	0.1483
fair/poor health	-0.21750	0.1559	-0.25813	0.1744
county violent crime rate	-0.13954	0.1165	-0.15338	0.1282
state no. of SAT facilities	-0.41756***	0.1264	-0.53476***	0.1437
time trend	0.07161	0.1675	0.08397	0.1800
time trend ²	-0.01481	0.0198	-0.02010	0.0213
time trend ³	0.00089	0.0007	0.00120	0.0007
constant	-1.44846**	0.5459	-4.29500***	0.7888
Permanent UH				
type 2			0.66412***	0.2356
type 3			0.46927*	0.2538
type 4			0.43634	0.4017
Time-varying UH				
type 2			2.55746***	0.4784
type 3			2.55478***	0.7071
type 4			4.31108***	0.5170

Note: Permanent and Time-varying UH are relative to type 1, which are normalized to 0.

Coefficients reported are log odds. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table OA.11: **Estimation of Cognitive Achievement: No UH vs With UH**

Variable	No UH		UH	
	Coefficient	Std. Dev.	Coefficient	Std. Dev.
mom with biodad	0.09071***	0.0304	0.16552***	0.0426
mom with anotherdad	0.11624**	0.0550	0.12257**	0.0585
mom employed	0.12578***	0.0343	0.11762**	0.0447
mom welfare	-0.08892**	0.0354	-0.11172**	0.0437
mom drinks	0.09976***	0.0271	0.01339	0.0585
mom smoke	-0.07257**	0.0312	-0.11379	0.0924
mom marijuana	0.11881	0.0794	0.05087	0.0877
mom convicted	-0.13225	0.0855	-0.17060*	0.0876
dad in jail	0.03705	0.0529	0.01365	0.0581
lagged cognitive score	0.31078***	0.0148	0.29800***	0.0175
white	0.26021***	0.0377	0.24166***	0.0397
hispanic	-0.03475	0.0368	-0.06898*	0.0404
other race	0.33676***	0.0800	0.29370***	0.0845
male	-0.11048***	0.0274	-0.11233***	0.0278
age	0.01241***	0.0025	0.01159***	0.0027
no. of kids in hh	-0.08053***	0.0103	-0.08025***	0.0105
grandparent present	0.12104***	0.0384	0.10830***	0.0392
depressed	0.05448	0.0646	0.05237	0.0646
withdrawn	-0.22577***	0.0625	-0.20991***	0.0630
attention problem	-0.13825**	0.0571	-0.13177**	0.0571
aggressive	-0.00835	0.0571	-0.00254	0.0569
defiant	-0.00764	0.0653	-0.01547	0.0648
daycare	0.11604***	0.0337	0.10493***	0.0344
county test score	0.00351**	0.0014	0.00346**	0.0014
time trend	0.09699	0.1073	0.12632	0.1092
time trend ²	-0.02816*	0.0173	-0.03124*	0.0175
time trend ³	0.00169**	0.0008	0.00181**	0.0009
constant	5.16944***	0.2518	5.03168***	0.2609
Permanent UH				
type 2			0.12306	0.1161
type 3			0.00012	0.1197
type 4			0.28355*	0.1703
Time-varying UH				
type 2			0.20221***	0.0753
type 3			-0.00195	0.1336
type 4			0.24597***	0.0958

Note: Family and Child permanent UH are relative to Family type 1 and Child type 1, respectively, which are normalized to 0. Coefficients reported are OLS estimates.

*** p<0.01, ** p<0.05, * p<0.1.

Table OA.12: Estimation of Ever Failed a Class: No UH vs With UH

Variable	No UH		UH	
	Coefficient	Std. Dev.	Coefficient	Std. Dev.
biodad always	-0.3707*	0.2167	-0.4133	0.2674
single early	0.1263	0.2207	0.1750	0.2554
biodad early	-0.1359	0.2128	-0.2243	0.2434
marry unstable	-0.1459	0.2201	-0.1453	0.2503
marry other	-0.1134	0.1982	-0.1705	0.2284
employ always	0.1642	0.3719	-0.2089	0.4359
employ early	-0.0986	0.4294	-0.0599	0.4687
employ late	0.2099	0.4393	0.0721	0.4927
employ early & late	0.3235	0.3627	0.1674	0.4006
welfare always	0.6470	0.6710	0.9497	0.6809
welfare early	0.3254**	0.1520	0.3820**	0.1780
welfare late	0.2554	0.2288	0.4170	0.2652
welfare early & late	0.2407	0.1798	0.3668*	0.2097
smoke always	0.3682	0.2331	0.7206	0.7170
smoke early	0.2631	0.2954	0.4925	0.4097
smoke late	0.2297	0.2221	0.4261	0.4056
smoke early & late	0.2281	0.1562	0.5847	0.6597
drink always	-0.2331	0.3215	-0.6010	0.5476
drink early	-0.0424	0.3307	0.1516	0.3743
drink late	0.0496	0.1872	0.0571	0.2410
drink early & late	-0.0990	0.1727	-0.2880	0.3429
marijuana ever	0.3520	0.2137	0.2045	0.2720
convicted early	0.3849	0.5742	0.4406	0.7527
convicted late	0.4472*	0.2396	0.4204	0.2767
jail early	-0.2290	0.1917	-0.2201	0.2238
jail late	0.0102	0.1748	0.0288	0.2181
lagged cognitive score	-0.1791**	0.0824	-0.2107**	0.0943
WJ Math	-0.0380***	0.0140	-0.0453***	0.0161
WJ English	0.0109	0.0157	0.0138	0.0176
male	0.0915	0.1146	0.0867	0.1297
white	-0.1666	0.1741	-0.2105	0.2064
hispanic	0.4045***	0.1524	0.4146**	0.1880
other race	-0.1625	0.3337	-0.2130	0.4080
mum less than high sch	-0.2704	0.1958	-0.3213	0.2263
mum some college	-0.2000	0.1585	-0.2182	0.1798
mum graduate	-0.3144*	0.1918	-0.2444	0.2206
peer drinks	-0.0241	0.1528	-0.0521	0.1710
peer smokes	0.2630	0.1687	0.2218	0.1892
peer marijuana	0.3870***	0.1444	0.4675***	0.1634
peer study more	-0.1575	0.1311	-0.1791	0.1522
peer likely grad college	-0.1289	0.1241	-0.1474	0.1416
public school	0.1251	0.1157	0.1636	0.1341
ADHD	0.3546	0.2050	0.3909*	0.2232
disobedient	0.7926***	0.1520	0.8650***	0.1707
cheats	0.1898	0.1372	0.2227	0.1539
school activities monthly	-0.3355**	0.1665	-0.3744**	0.1901
school activities weekly	-0.1911	0.1788	-0.2979	0.2091
school activities often	-0.0723	0.2056	-0.0643	0.2314
religious activity monthly	-0.2659	0.1662	-0.2796	0.1876
religious activity weekly	-0.2565	0.1559	-0.3386*	0.1811
religious activity often	-0.0525	0.2104	-0.0274	0.2356
county test score	0.0027	0.0059	0.0043	0.0068
constant	1.8195***	0.6780	2.9725***	0.8810
Permanent UH				
type 2			0.3019	0.5012
type 3			-0.5526	0.7497
type 4			0.2890	1.0110
Time-varying UH				
type 2			-1.1458***	0.3972
type 3			1.0756	0.8981
type 4			-0.4504	0.4385

Note: Permanent and Time-varying UH are relative to type 1, which are normalized to 0.
Coefficients reported are log odds. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table OA.13: Estimation of Unlikely to Graduate College: No UH vs With UH

Variable	No UH		UH	
	Coefficient	Std. Dev.	Coefficient	Std. Dev.
biodad always	-0.3611	0.2266	-0.3925	0.2582
single early	0.1912	0.2278	0.2447	0.2555
biodad early	-0.1687	0.2219	-0.2161	0.2416
marry unstable	0.1116	0.2242	0.1420	0.2435
marry other	-0.2143	0.2070	-0.2720	0.2220
employ always	0.1868	0.3802	-0.0390	0.4191
employ early	0.0406	0.4441	0.0857	0.4534
employ late	0.3217	0.4711	0.1813	0.4927
employ early & late	0.5308	0.3712	0.4339	0.3826
welfare always	0.1416	0.5586	0.4287	0.5741
welfare early	-0.0534	0.1647	-0.0523	0.1786
welfare late	0.1224	0.2332	0.2304	0.2568
welfare early & late	0.0438	0.1886	0.1132	0.2076
smoke always	-0.1645	0.2344	0.3736	0.8168
smoke early	0.2305	0.3215	0.5434	0.4119
smoke late	-0.1848	0.2421	0.1478	0.3872
smoke early & late	-0.1772	0.1614	0.3546	0.7442
drink always	0.5518*	0.3367	0.7243	0.4988
drink early	0.0989	0.3492	0.3519	0.3892
drink late	0.4051**	0.2023	0.5647**	0.2415
drink early & late	0.3969**	0.1877	0.5698*	0.3191
marijuana ever	0.3610*	0.2182	0.3026	0.2525
convicted early	-0.3083	0.6265	-0.4428	0.6838
convicted late	0.4002	0.2509	0.3762	0.2637
jail early	-0.2054	0.1984	-0.1399	0.2178
jail late	-0.2123	0.1882	-0.1834	0.2193
lagged cognitive score	0.0645	0.0840	0.0732	0.0930
WJ Math	-0.0242*	0.0135	-0.0285*	0.0146
WJ English	-0.0247	0.0154	-0.0247	0.0168
male	0.3647***	0.1229	0.3937***	0.1326
white	0.5041***	0.1835	0.5091**	0.2023
hispanic	0.7089***	0.1587	0.7691***	0.1821
other race	0.5456	0.3399	0.6239	0.3750
mum less than high sch	0.1288	0.1996	0.1192	0.2197
mum some college	-0.1999	0.1610	-0.2300	0.1749
mum graduate	-0.4855**	0.2081	-0.4273*	0.2302
peer drinks	0.2397	0.1593	0.2539	0.1739
peer smokes	0.3320*	0.1743	0.2863	0.1895
peer marijuana	0.1817	0.1527	0.2269	0.1648
peer study more	-0.1173	0.1330	-0.1330	0.1481
peer likely grad college	-0.5459***	0.1328	-0.5753***	0.1427
public school	0.0346	0.1232	0.0652	0.1350
ADHD	0.4132**	0.1937	0.4659**	0.2072
disobedient	0.4332***	0.1541	0.4434***	0.1651
cheats	0.3779***	0.1394	0.4096***	0.1504
school activities monthly	-0.3996**	0.1893	-0.4147**	0.1985
school activities weekly	0.0052	0.1872	-0.0602	0.2045
school activities often	-0.3019	0.2319	-0.2987	0.2449
religious activity monthly	-0.3303*	0.1765	-0.3450*	0.1932
religious activity weekly	-0.3370*	0.1737	-0.3828**	0.1860
religious activity often	-0.1445	0.2243	-0.1639	0.2367
county test score	0.0060	0.0064	0.0077	0.0070
constant	-0.6191	0.6892	-0.0530	0.8721
Permanent UH				
type 2			-0.2517	0.4003
type 3			-0.9369	0.7866
type 4			-0.4418	0.9939
Time-varying UH				
type 2			-0.8687**	0.4418
type 3			0.7316	0.7723
type 4			-0.4234	0.5338

Note: Permanent and Time-varying UH are relative to type 1, which are normalized to 0.
Coefficients reported are log odds. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table OA.14: Estimation of Ever Suspended from School: No UH vs With UH

Variable	No UH		UH	
	Coefficient	Std. Dev.	Coefficient	Std. Dev.
biodad always	-0.1092	0.2660	0.0778	0.3316
single early	0.5726**	0.2471	0.7288**	0.3014
biodad early	0.1948	0.2521	0.1732	0.2961
marry unstable	0.4686*	0.2489	0.5990**	0.2921
marry other	0.3013	0.2240	0.3395	0.2702
employ always	-0.1866	0.3948	-0.3424	0.4532
employ early	-0.1437	0.4398	-0.1740	0.4758
employ late	0.3319	0.4285	0.2709	0.4617
employ early & late	0.4538	0.3590	0.4727	0.3842
welfare always	0.6337	0.6414	0.9793	0.6704
welfare early	0.1713	0.1691	0.2175	0.1983
welfare late	0.2700	0.2302	0.3660	0.2689
welfare early & late	0.2090	0.1839	0.3127	0.2150
smoke always	0.0568	0.2591	0.4521	0.7827
smoke early	0.4221	0.3186	0.7201*	0.4142
smoke late	0.0047	0.2393	0.3279	0.3999
smoke early & late	-0.0411	0.1600	0.4866	0.7625
drink always	0.0483	0.3535	0.2262	0.5316
drink early	-0.1817	0.3642	0.0543	0.4115
drink late	0.2316	0.2013	0.3656	0.2523
drink early & late	0.2437	0.1859	0.4007	0.3288
marijuana ever	0.1075	0.2201	0.0432	0.2655
convicted early	0.2222	0.6640	0.0591	0.7353
convicted late	0.1864	0.2558	0.1325	0.2776
jail early	-0.0334	0.2075	0.0336	0.2436
jail late	0.0413	0.1824	0.0616	0.2150
lagged cognitive score	0.0243	0.0884	0.0164	0.1002
WJ Math	-0.0368***	0.0135	-0.0440***	0.0148
WJ English	-0.0046	0.0154	-0.0046	0.0174
male	0.4680***	0.1226	0.4881***	0.1374
white	0.4824***	0.1925	0.4899**	0.2070
hispanic	0.6525***	0.1643	0.6940***	0.1908
other race	0.5137	0.3394	0.5846	0.3806
mum less than high sch	0.1638	0.2018	0.1680	0.2256
mum some college	-0.1348	0.1655	-0.1588	0.1820
mum graduate	-0.4001*	0.2092	-0.3308	0.2344
peer drinks	0.1624	0.1607	0.1726	0.1811
peer smokes	0.2222	0.1728	0.1882	0.1958
peer marijuana	0.1046	0.1527	0.1490	0.1718
peer study more	-0.0965	0.1318	-0.1105	0.1511
peer likely grad college	-0.4515***	0.1328	-0.4756***	0.1513
public school	0.1002	0.1235	0.1170	0.1415
ADHD	0.3915**	0.1950	0.4489**	0.2160
disobedient	0.3247**	0.1555	0.3408*	0.1785
cheats	0.3300**	0.1420	0.3533**	0.1614
school activities monthly	-0.4400**	0.1906	-0.4560**	0.2092
school activities weekly	0.0008	0.1851	-0.0567	0.2066
school activities often	-0.1997	0.2328	-0.1831	0.2549
religious activity monthly	-0.3576**	0.1762	-0.3744*	0.2002
religious activity weekly	-0.3751**	0.1756	-0.4167**	0.1975
religious activity often	-0.0435	0.2255	-0.0301	0.2537
county test score	-0.0151**	0.0072	-0.0161*	0.0084
constant	-2.7201***	0.8515	-2.2558**	1.0848
Permanent UH				
type 2			0.2428	0.5428
type 3			0.2407	0.8483
type 4			1.6706	1.0956
Time-varying UH				
type 2			-0.7334	0.4951
type 3			1.6931**	0.7963
type 4			0.6311	0.5274

Table OA.15: Estimation of Ever Drunk Alcohol: No UH vs With UH

Variable	No UH		UH	
	Coefficient	Std. Dev.	Coefficient	Std. Dev.
biodad always	0.7778**	0.3857	1.4154***	0.5323
single early	0.8850**	0.3727	1.2575***	0.4804
biodad early	0.4992	0.3689	0.4410	0.4421
marry unstable	0.7201*	0.3713	1.1211**	0.4720
marry other	0.6558*	0.3489	0.9707**	0.4329
employ always	1.1256*	0.6409	1.1133	0.8458
employ early	0.0168	0.8129	0.1204	0.9612
employ late	0.8844	0.8029	0.9635	1.0507
employ early & late	1.0216	0.6390	1.1327	0.8086
welfare always	0.6486	0.5433	0.6029	0.8542
welfare early	-0.2522	0.1663	-0.1742	0.2030
welfare late	0.2285	0.2382	0.2948	0.2882
welfare early & late	-0.1106	0.1840	-0.0150	0.2279
smoke always	-0.2334	0.2774	-0.2249	0.3285
smoke early	-0.2317	0.3495	-0.2200	0.4231
smoke late	-0.3011	0.2732	-0.1593	0.3195
smoke early & late	-0.3144*	0.1724	-0.3744*	0.2004
drink always	0.7817	0.5180	1.4363	0.8887
drink early	0.4479	0.5156	0.8311	0.6463
drink late	1.1679***	0.3318	1.5932***	0.4648
drink early & late	1.0933***	0.2607	1.4898***	0.4119
marijuana ever	0.2927	0.2540	0.2513	0.2905
convicted early	-0.5061	0.7138	-0.6661	0.7481
convicted late	-0.0981	0.2893	-0.1496	0.3212
jail early	-0.0324	0.2509	-0.0251	0.2877
jail late	0.2872	0.2282	0.3073	0.2749
lagged cognitive score	0.1047	0.0759	0.1272	0.0896
WJ Math	-0.0154	0.0144	-0.0220	0.0156
WJ English	0.0110	0.0159	0.0160	0.0179
male	1.0745***	0.1973	1.1028***	0.2159
white	0.4544**	0.2116	0.4786**	0.2358
hispanic	0.4542**	0.1884	0.5218**	0.2289
other race	0.5007	0.3626	0.5603	0.4033
mum less than high sch	-0.3441	0.2249	-0.3532	0.2518
mum some college	-0.2202	0.1713	-0.2375	0.1900
mum graduate	-0.3593	0.2536	-0.3284	0.2864
peer drinks	0.5517**	0.2376	0.6099**	0.2710
peer smokes	0.5899***	0.2128	0.6514***	0.2472
peer marijuana	0.3080	0.2060	0.3444	0.2348
peer study more	0.3238	0.2163	0.3410	0.2402
peer likely grad college	-1.2901***	0.2032	-1.3574***	0.2243
public school	0.0120	0.1349	0.0270	0.1510
ADHD	-0.0888	0.2287	-0.0489	0.2541
disobedient	0.2445	0.1798	0.2545	0.2089
cheats	0.0856	0.1619	0.0996	0.1891
school activities monthly	-0.5088**	0.2362	-0.5378**	0.2623
school activities weekly	-0.0294	0.2022	-0.0490	0.2304
school activities often	-0.4742*	0.2497	-0.4710*	0.2741
religious activity monthly	-0.3275	0.2110	-0.3437	0.2379
religious activity weekly	-0.1683	0.1975	-0.2002	0.2242
religious activity often	-0.2427	0.2388	-0.2188	0.2718
county test score	0.0088	0.0073	0.0108	0.0081
constant	-4.0633***	0.9456	-3.5065***	1.1923
Permanent UH				
type 2			0.6583	0.6928
type 3			0.0431	1.0724
type 4			-0.1486	1.3147
Time-varying UH				
type 2			-2.3851***	0.7067
type 3			0.5227	0.9251
type 4			0.7944	0.6315

Note: Permanent and Time-varying UH are relative to type 1, which are normalized to 0.
Coefficients reported are log odds. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table OA.16: Estimation of Ever Use Marijuana: No UH vs With UH

Variable	No UH		UH	
	Coefficient	Std. Dev.	Coefficient	Std. Dev.
biodad always	-0.5750*	0.3290	0.3344	0.6779
single early	0.2595	0.3092	0.6710	0.5718
biodad early	0.0919	0.2890	-0.1070	0.5031
marry unstable	0.0740	0.3100	0.2996	0.5734
marry other	0.1248	0.2654	0.4189	0.4911
employ always	1.1185*	0.6499	1.6528	1.0859
employ early	0.3880	0.7505	1.2181	1.1711
employ late	0.6983	0.7237	1.3565	1.2881
employ early & late	1.1668*	0.6394	2.0946**	1.0592
welfare always	0.7809	0.6391	0.4395	1.3761
welfare early	-0.0921	0.2350	-0.0566	0.3818
welfare late	0.1833	0.2990	0.4051	0.5209
welfare early & late	0.3353	0.2534	0.5027	0.4471
smoke always	0.7310**	0.3008	-0.3013	1.0995
smoke early	-0.4151	0.4827	-1.2940	1.0024
smoke late	0.1648	0.2986	-0.5458	0.6285
smoke early & late	0.1302	0.2224	-0.8846	0.9971
drink always	0.5520	0.4224	-0.1621	0.9463
drink early	-0.2147	0.5171	0.2523	0.8595
drink late	0.0187	0.2740	-0.1815	0.5508
drink early & late	0.1535	0.2498	-0.4587	0.6224
marijuana ever	0.2523	0.2753	0.1350	0.4802
convicted early	0.9706	0.6899	0.9787	1.5174
convicted late	0.3968	0.2710	-0.0912	0.4913
jail early	0.4050	0.2513	0.5062	0.4271
jail late	0.5270**	0.2316	-0.1622	0.5383
lagged cognitive score	-0.0429	0.1158	-0.2101	0.1866
WJ Math	0.0360*	0.0189	0.0695**	0.0336
WJ English	-0.0171	0.0204	-0.0399	0.0336
male	0.4097**	0.1718	0.7051**	0.2872
white	0.0837	0.2594	-0.2980	0.4592
hispanic	0.3999*	0.2139	0.3188	0.3827
other race	-0.1223	0.5767	-0.2549	1.0959
mum less than high sch	0.0002	0.2772	0.2523	0.4577
mum some college	-0.2195	0.2324	-0.1465	0.3881
mum graduate	-0.6000**	0.3037	-1.6152***	0.5949
peer drinks	0.8221***	0.1919	1.3288***	0.3879
peer smokes	-0.0096	0.1979	-0.2165	0.3840
peer marijuana	2.2083***	0.1956	4.1674***	0.5806
peer study more	0.0195	0.1735	-0.0262	0.3112
peer likely grad college	-0.2169	0.1846	-0.3691	0.3458
public school	0.0903	0.1733	0.3277	0.3215
ADHD	-0.2006	0.2695	-0.5361	0.4357
disobedient	0.4234**	0.2000	0.5839*	0.3415
cheats	0.6110***	0.1796	0.9979***	0.3119
school activities monthly	-0.0857	0.2670	0.0505	0.4278
school activities weekly	0.1057	0.2502	-0.1436	0.4981
school activities often	-0.6999*	0.3634	-1.1448**	0.5681
religious activity monthly	-0.7956***	0.2539	-1.2102**	0.5258
religious activity weekly	-0.3894	0.2590	-0.4933	0.4055
religious activity often	-0.1347	0.3516	0.1922	0.6099
county test score	-0.0149*	0.0080	-0.0175	0.0136
constant	-5.2435***	1.0300	-7.9961***	2.0216
Permanent UH				
type 2			1.5523*	0.9138
type 3			2.0820*	1.2318
type 4			3.0540*	1.6541
Time-varying UH				
type 2			-2.1733***	0.6946
type 3			2.1643*	1.2602
type 4			3.7816***	0.9665

Note: Permanent and Time-varying UH are relative to type 1, which are normalized to 0.
Coefficients reported are log odds. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table OA.17: Estimation of Ever Stopped by Police: No UH vs With UH

Variable	No UH		UH	
	Coefficient	Std. Dev.	Coefficient	Std. Dev.
biodad always	-0.0037	0.2465	0.1535	0.2941
single early	0.2332	0.2501	0.2548	0.2928
biodad early	0.4293*	0.2265	0.4482*	0.2613
marry unstable	0.4982**	0.2441	0.6304**	0.2759
marry other	0.0435	0.2247	0.0930	0.2546
employ always	-0.5323	0.3889	-0.7885*	0.4392
employ early	-0.2378	0.4931	-0.1920	0.5372
employ late	0.1578	0.4634	0.1784	0.5086
employ early & late	-0.3541	0.3752	-0.4377	0.4089
welfare always	2.0889***	0.5632	2.4354***	0.6306
welfare early	-0.0424	0.1786	-0.0187	0.1972
welfare late	0.2375	0.2419	0.3066	0.2732
welfare early & late	0.2743	0.2023	0.3906*	0.2274
smoke always	-0.0463	0.2579	-1.3037	0.9523
smoke early	-0.1730	0.3706	-0.6270	0.5737
smoke late	0.1149	0.2518	-0.4442	0.4628
smoke early & late	0.1769	0.1744	-0.8996	0.8565
drink always	-0.6487	0.4085	-1.5363**	0.6809
drink early	-0.1939	0.4121	-0.1324	0.4740
drink late	0.0537	0.2165	-0.1524	0.2981
drink early & late	0.1653	0.1998	-0.3130	0.4042
marijuana ever	0.0131	0.2317	-0.2524	0.2860
convicted early	-0.2490	0.6420	-0.3755	0.6798
convicted late	0.0782	0.2632	-0.1407	0.3078
jail early	0.2903	0.2135	0.2856	0.2424
jail late	0.4046**	0.1854	0.2729	0.2292
lagged cognitive score	0.0063	0.0906	-0.0358	0.1010
WJ Math	0.0315**	0.0141	0.0334**	0.0152
WJ English	0.0067	0.0169	0.0059	0.0180
male	1.1302***	0.1375	1.2106***	0.1546
white	-0.1532	0.1965	-0.1911	0.2235
hispanic	-0.4021**	0.1816	-0.6066***	0.2243
other race	-0.2013	0.3842	-0.3040	0.4304
mum less than high sch	-0.0538	0.2241	-0.0733	0.2514
mum some college	0.1571	0.1862	0.1779	0.2011
mum graduate	-0.0968	0.2248	-0.2031	0.2557
peer drinks	0.4155**	0.1746	0.4250**	0.1898
peer smokes	0.1075	0.1813	0.0714	0.1971
peer marijuana	0.8542***	0.1596	0.9802***	0.1803
peer study more	-0.0613	0.1445	-0.0867	0.1593
peer likely grad college	-0.3795***	0.1435	-0.4166***	0.1588
public school	0.0131	0.1336	0.0505	0.1476
ADHD	-0.1673	0.2157	-0.1901	0.2334
disobedient	0.5981***	0.1637	0.6589***	0.1848
cheats	0.0991	0.1533	0.0929	0.1684
school activities monthly	-0.2219	0.2057	-0.2091	0.2262
school activities weekly	-0.0026	0.2076	-0.0607	0.2228
school activities often	-0.0201	0.2281	-0.0295	0.2488
religious activity monthly	-0.2517	0.2029	-0.1952	0.2196
religious activity weekly	0.0188	0.1874	0.0328	0.2010
religious activity often	0.0194	0.2464	0.1703	0.2801
county test score	-0.0143**	0.0067	-0.0145*	0.0073
constant	-3.4905***	0.7521	-2.5471***	0.8966
Permanent UH				
type 2			0.7398	0.5209
type 3			1.1806	0.9086
type 4			2.2373*	1.2796
Time-varying UH				
type 2			-1.3083**	0.5199
type 3			-0.3511	0.7667
type 4			0.1177	0.5180

Note: Permanent and Time-varying UH are relative to type 1, which are normalized to 0.
Coefficients reported are log odds. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table OA.18: Estimation of Ever Had Sex: No UH vs With UH

Variable	No UH		UH	
	Coefficient	Std. Dev.	Coefficient	Std. Dev.
biodad always	-0.4700	0.3203	0.3303	0.4402
single early	0.3087	0.2758	0.6568	0.4290
biodad early	0.3172	0.2790	0.4342	0.3942
marry unstable	0.0633	0.2973	0.3608	0.4266
marry other	0.2989	0.2554	0.6819*	0.3623
employ always	0.5124	0.6280	0.1615	0.7305
employ early	0.0760	0.7157	-0.1571	0.8211
employ late	-0.9073	0.7927	-1.4817	1.0091
employ early & late	0.5634	0.6103	0.4795	0.6579
welfare always	1.6861***	0.5762	2.1805***	0.7609
welfare early	0.1389	0.2327	0.2552	0.3061
welfare late	-0.0056	0.3247	0.0317	0.4658
welfare early & late	0.5633**	0.2443	0.8210**	0.3376
smoke always	0.2854	0.2840	-0.0194	1.0728
smoke early	-1.0980*	0.6505	-2.0933**	0.9968
smoke late	0.2532	0.2981	-0.1410	0.5043
smoke early & late	-0.0202	0.2187	-0.1723	0.9908
drink always	-0.2078	0.4302	-1.2529	0.8722
drink early	0.2170	0.4525	0.5633	0.6165
drink late	-0.0283	0.2597	-0.4360	0.4882
drink early & late	0.0390	0.2414	-0.6632	0.5936
marijuana ever	0.2852	0.2637	0.1439	0.3587
convicted early	0.9257	0.7639	1.0461	1.1709
convicted late	-0.2630	0.2940	-0.7241*	0.4179
jail early	0.4378*	0.2282	0.3349	0.3097
jail late	0.0021	0.2223	-0.7924**	0.3829
lagged cognitive score	-0.1243	0.1131	-0.2520	0.1566
WJ Math	0.0401*	0.0202	0.0643**	0.0276
WJ English	-0.0048	0.0227	-0.0145	0.0285
male	0.9078***	0.1793	1.0895***	0.2307
white	-0.3673	0.2541	-0.9115***	0.3447
hispanic	-0.5957**	0.2442	-1.0495***	0.3355
other race	-0.8290	0.5982	-1.5150	1.0603
mum less than high sch	0.2214	0.2691	0.3765	0.3562
mum some college	0.0532	0.2260	0.1866	0.2947
mum graduate	-0.5300*	0.3089	-1.0033**	0.4273
peer drinks	0.4576**	0.1961	0.5399*	0.2857
peer smokes	0.1612	0.2138	0.0544	0.3000
peer marijuana	1.0804***	0.1927	1.5807***	0.3007
peer study more	-0.2582	0.1740	-0.4135*	0.2404
peer likely grad college	-0.2610	0.1891	-0.3718	0.2573
public school	-0.0578	0.1655	0.1227	0.2259
ADHD	0.1279	0.2436	0.1419	0.3254
disobedient	0.2786	0.1860	0.3458	0.2625
cheats	0.6187***	0.1796	0.8353***	0.2635
school activities monthly	-0.0844	0.2566	-0.0606	0.3474
school activities weekly	-0.4025	0.3100	-0.7782*	0.4065
school activities often	0.0648	0.2855	0.1554	0.3816
religious activity monthly	-0.7101**	0.2966	-0.7683**	0.3721
religious activity weekly	0.0279	0.2471	0.1073	0.3289
religious activity often	-0.2194	0.3396	-0.0872	0.4590
county test score	-0.0080	0.0086	-0.0027	0.0117
constant	-3.7392***	1.0078	-4.1144***	1.4697
Permanent UH				
type 2			1.5682**	0.7632
type 3			1.1106	1.1124
type 4			1.3263	1.2954
Time-varying UH				
type 2			-1.5363**	0.6776
type 3			1.2438	0.8812
type 4			2.2730***	0.7887

Note: Permanent and Time-varying UH are relative to type 1, which are normalized to 0.
Coefficients reported are log odds. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

3 Results from Alternative Models

3.1 Results for Behavior Summation Model

Table OA.19: Summary Statistics for Model Fit – Behavior Summation Model

	Observed		Simulated	
	Mean	Std. Dev.	Mean	Std. Dev.
<i>Parental Behaviors</i>				
Married/cohabit status				
single	0.4501	0.4975	0.4148	0.4927
with biodad	0.4838	0.4997	0.5282	0.4992
with anotherman	0.0662	0.2485	0.0571	0.2320
Employed	0.7820	0.4129	0.7831	0.4122
Welfare Receipt	0.2028	0.4021	0.2075	0.4055
Substance Use				
Drinking	0.4757	0.4994	0.4756	0.4994
Smoking	0.2430	0.4289	0.2314	0.4217
Marijuana use	0.0254	0.1573	0.0245	0.1547
Conviction	0.0159	0.1251	0.0262	0.1598
Jail	0.0764	0.2656	0.0847	0.2784
<i>Child</i>				
Cognitive score	80.0119	11.1887	80.0243	11.0789
<i>Teenage outcomes</i>				
Ever failed class	0.4619	0.4985	0.4609	0.4985
Unlikely to graduate college	0.3251	0.4685	0.3140	0.4641
Ever suspended from school	0.2755	0.4468	0.2813	0.4496
Ever drunk alcohol	0.1300	0.3363	0.1313	0.3377
Ever used marijuana	0.1777	0.3823	0.1806	0.3847
Ever stopped by police	0.2570	0.4370	0.2591	0.4381
Ever had sex	0.1505	0.3575	0.1512	0.3583

Table OA.20: Probabilities Weights – Behavior Summation Model

Type	Permanent Heterogeneity		Time-varying Heterogeneity	
	Probability	Estimated Coefficient	Probability	Estimated Coefficient
1	0.2305	0.0000 (0.0000)	0.1630	0.0000 (0.0000)
2	0.1656	-0.3304 (0.3713)	0.5032	1.1272 (0.2076)
3	0.4109	0.5780 (0.4391)	0.1611	-0.0118 (0.1904)
4	0.1930	-0.1777 (0.3769)	0.1727	0.0574 (0.1880)

Notes: Standard errors are in parentheses.

Figure OA.1: Model Fit for Dynamic Behaviors and Cognitive Achievement – Behavior Summation Model

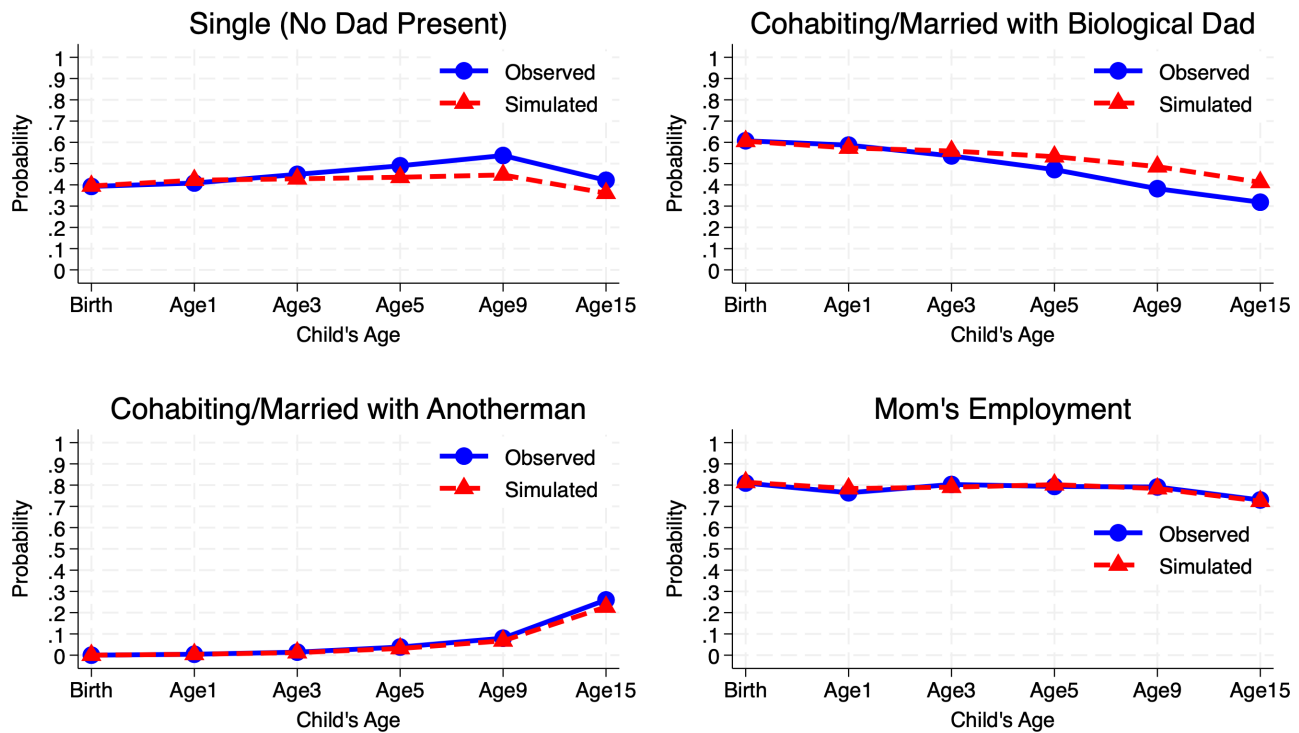
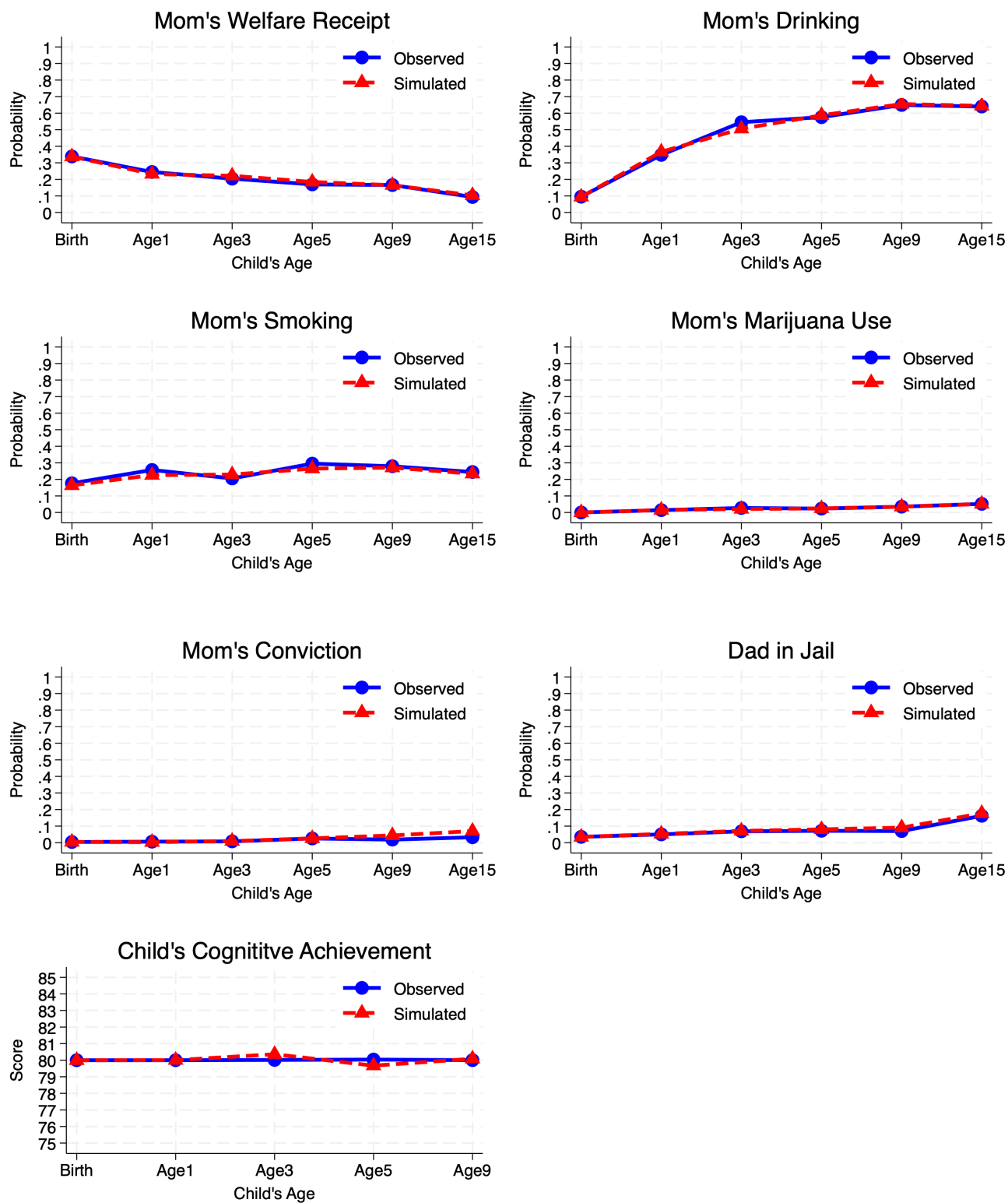


Figure OA.1: Model Fit for Dynamic Behaviors and Cognitive Achievement – Behavior Summation Model (continued)



3.2 Results for Contemporaneous Parental Behavior Model

Table OA.21: **Summary Statistics for Model Fit – Contemporaneous Parental Behavior Model**

	Observed		Simulated	
	Mean	Std. Dev.	Mean	Std. Dev.
<i>Parental Behaviors</i>				
Married/cohabit status				
single	0.4501	0.4975	0.4175	0.4932
with biodad	0.4838	0.4997	0.5256	0.4993
with anotherman	0.0662	0.2485	0.0567	0.2316
Employed	0.7820	0.4129	0.7827	0.4124
Welfare Receipt	0.2028	0.4021	0.2070	0.4052
Substance Use				
Drinking	0.4757	0.4994	0.4769	0.4995
Smoking	0.2430	0.4289	0.2318	0.4220
Marijuana use	0.0254	0.1573	0.0253	0.1569
Conviction	0.0159	0.1251	0.0180	0.1331
Jail	0.0764	0.2656	0.0854	0.2795
<i>Child</i>				
Cognitive score	80.0119	11.1887	80.0254	11.0778
<i>Teenage outcomes</i>				
Ever failed class	0.4619	0.4985	0.4555	0.4980
Unlikely to graduate college	0.3251	0.4685	0.3135	0.4639
Ever suspended from school	0.2755	0.4468	0.2794	0.4487
Ever drunk alcohol	0.1300	0.3363	0.1363	0.3431
Ever used marijuana	0.1777	0.3823	0.1799	0.3841
Ever stopped by police	0.2570	0.4370	0.2584	0.4378
Ever had sex	0.1505	0.3575	0.1514	0.3584

Table OA.22: Probabilities Weights – Contemporaneous Parental Behavior Model

Type	Permanent Heterogeneity		Time-varying Heterogeneity	
	Probability	Estimated Coefficient	Probability	Estimated Coefficient
1	0.2818	0.0000 (0.0000)	0.1593	0.0000 (0.0000)
2	0.1635	−0.5444 (0.2689)	0.1520	−0.0466 (0.1988)
3	0.3660	0.2613 (0.2893)	0.5209	1.1849 (0.2127)
4	0.1887	−0.4010 (0.2121)	0.1677	0.0517 (0.1974)

Notes: Standard errors are in parentheses.

Figure OA.2: Model Fit for Dynamic Behaviors and Cognitive Achievement – Contemporaneous Parental Behavior Model

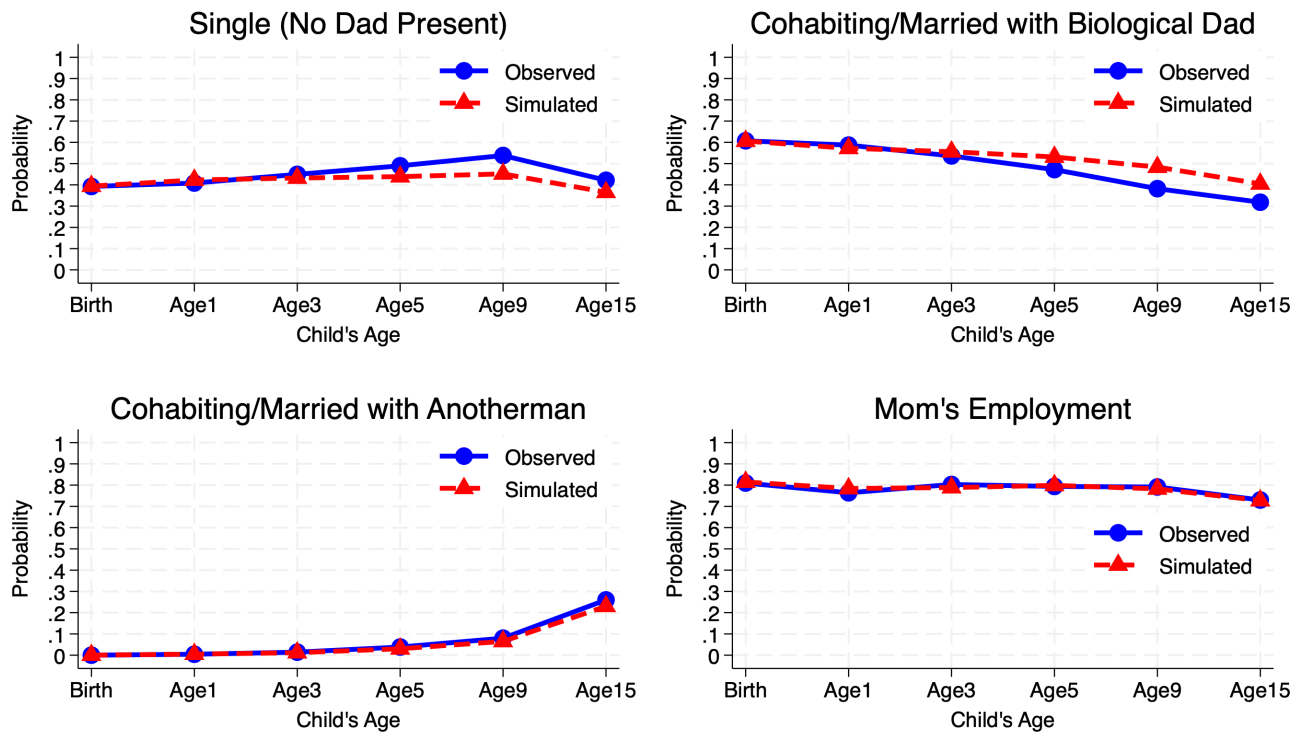


Figure OA.2: Model Fit for Dynamic Behaviors and Cognitive Achievement – Contemporaneous Parental Behavior Model (continued)

